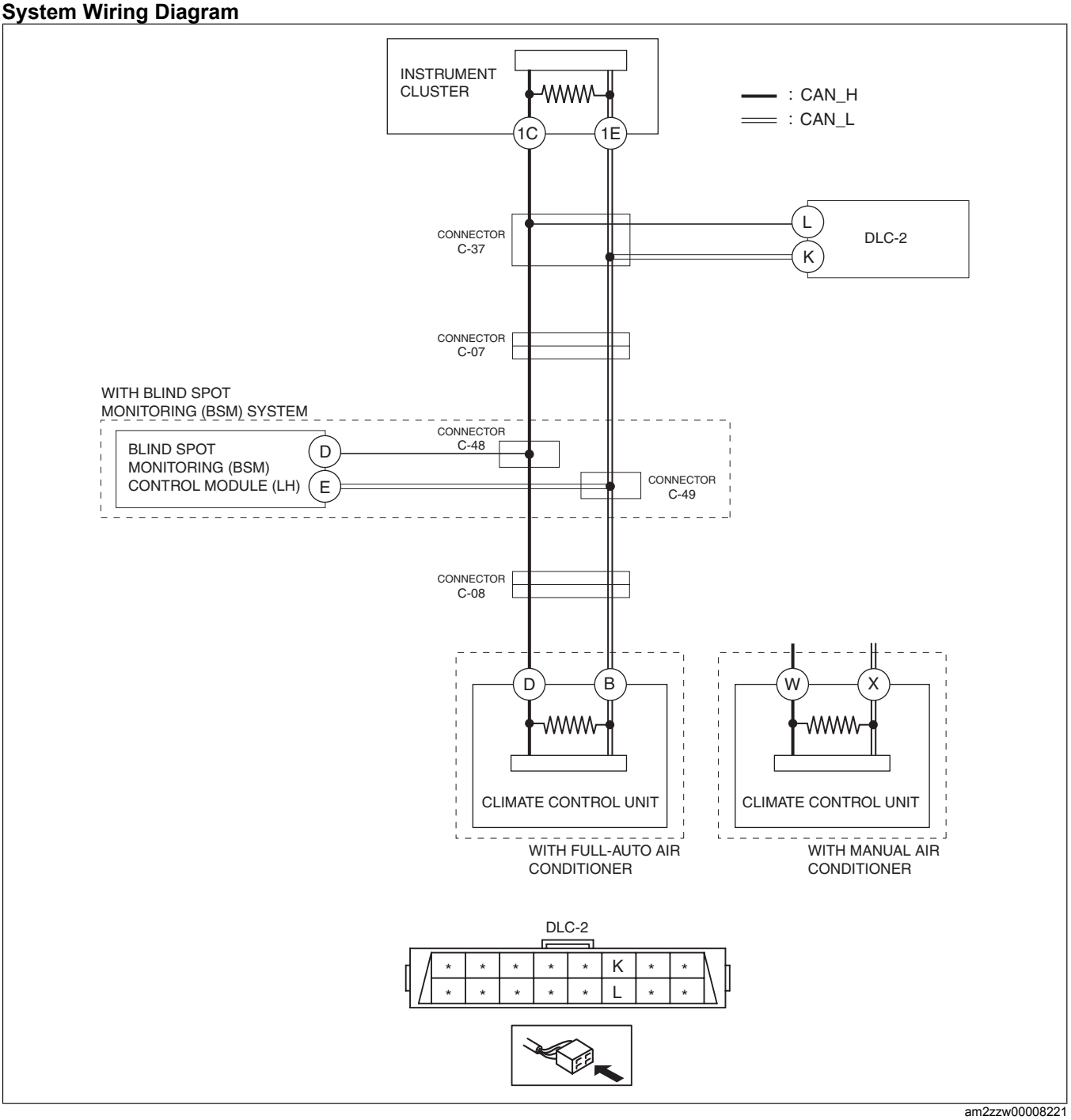
DETERMINING SHORT TO POWER SUPPLY LOCATION (MS-CAN) [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)]

id100220001000

Caution

- **Perform the following malfunction diagnosis only when it is diagnosed with a short to the power supply by CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW. (See CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)].)**

System Wiring Diagram



am2zzw00008221

Determination procedure

Caution

- **When disconnecting the connector, verify that there is no looseness, damage, deformation, corrosion, or poor connection of the connector terminals.**

- **When inspecting the DLC-2, touch it with a paper clip or similar thin pin without directly inserting a tester into the terminals.**
- **Disconnect the negative battery cable before performing any work that requires handling of connectors.**

Step	Inspection		Action
1	**INSPECT FOR SHORT TO POWER SUPPLY BETWEEN CONNECTOR C-37 AND INSTRUMENT CLUSTER** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the connector C-37. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at instrument cluster terminals 1C and 1E. • Is the voltage between **1.5—3.5 V**?	Yes	Go to Step 3.
		No	Go to the next step.
2	**INSPECT INSTRUMENT CLUSTER FOR SHORT TO POWER SUPPLY** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the instrument cluster connector. • Connect the connector C-37. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals L and K. • Is the voltage between **1.5—3.5 V**?	Yes	Replace the instrument cluster because there is a short to the power supply in the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)
		No	Repair or replace the wiring harness between the instrument cluster and connector C-37 because the wiring harness is shorted to the power supply.
3	**INSPECT FOR SHORT TO POWER SUPPLY BETWEEN CONNECTORS C-37 AND DLC-2** • Measure the voltage at DLC-2 terminals L and K. • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between DLC-2 and connector C-37 because the wiring harness is shorted to the power supply.
4	**INSPECT FOR SHORT TO POWER SUPPLY BETWEEN CONNECTOR C-07 AND CONNECTOR C-37** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the connector C-07. • Connect the connector C-37. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals L and K. • Is the voltage between **1.5—3.5 V**?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between connector C-37 and connector C-07 because the wiring harness is shorted to the power supply.

Step	Inspection		Action
5	**INSPECT FOR SHORT TO POWER SUPPLY BETWEEN CONNECTORS C-48 AND C49 AND CONNECTOR C-07** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Connect the connector C-07. • Disconnect the connectors C-48 and C-49. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals L and K. • Is the voltage between **1.5—3.5 V**?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between connector C-07 and connectors C-48 and C-49 because the wiring harness is shorted to the power supply.
6	**INSPECT FOR SHORT TO POWER SUPPLY BETWEEN BLIND SPOT MONITORING (BSM) CONTROL MODULE (LH) AND CONNECTORS C-48 AND C49** • Measure the voltage at blind spot monitoring (BSM) control module (LH) terminals D and E. • Is the voltage between **1.5—3.5 V**?	Yes	Go to Step 8.
		No	Go to the next step.
7	**INSPECT BLIND SPOT MONITORING (BSM) CONTROL MODULE (LH) FOR SHORT TO POWER SUPPLY** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the blind spot monitoring (BSM) control module (LH) connector. • Connect the connectors C-48 and C-49. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals L and K. • Is the voltage between **1.5—3.5 V**?	Yes	Replace the blind spot monitoring (BSM) control module (LH) because there is a short to power supply in the blind spot monitoring (BSM) control module (LH). (See BLIND SPOT MONITORING (BSM) CONTROL MODULE REMOVAL/INSTALLATION.)
		No	Repair or replace the wiring harness between the blind spot monitoring (BSM) control module (LH) and connectors C-48 and C-49 because the wiring harness is shorted to the power supply.
8	**INSPECT FOR SHORT TO POWER SUPPLY BETWEEN CONNECTORS C-48 AND C-49 AND CONNECTOR C-08** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Connect the connectors C-48 and C-49. • Disconnect the connector C-08. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals L and K. • Is the voltage between **1.5—3.5 V**?	Yes	Go to the next step.
		No	Repair or replace the wiring harness between the connector C-08 and connectors C-48 and C-49 because the wiring harness is shorted to the power supply.

Step	Inspection		Action
9	**INSPECT CLIMATE CONTROL UNIT FOR SHORT TO POWER SUPPLY** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the climate control unit connector. • Connect the connector C-08. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). • Measure the voltage at DLC-2 terminals L and K. • Is the voltage between **1.5—3.5 V**?	Yes	Replace the climate control unit because there is a short to the power supply in the climate control unit. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [MANUAL AIR CONDITIONER].)
		No	Repair or replace the wiring harness between the climate control unit and connector C-08 because the wiring harness is shorted to the power supply.